

Timothy Asady-Kany

US Citizen | tasadykany@gmail.com | 571-275-4716 | Orlando, FL | [LinkedIn](#) | [Portfolio](#)

EDUCATION

University of Central Florida | B.S. Aerospace Engineering

Class of 2028

GPA: 3.34/4.00

Lake Braddock Secondary School | High School Advanced Diploma

2018-2024

GPA: 3.75/4.00

SKILLS AND COURSEWORK

- **Experience with:** SolidWorks, Fusion, OnShape, OpenRocket, Microsoft Office (Excel, Outlook, PowerPoint Word), GitHub
- **Programming skills:** C, Python, MATLAB, HTML, CSS
- **Relevant Coursework:** Structures and Properties of Aerospace Materials, Mechanical Engineering Problem Solving
- **Languages:** English (Native), French (Fluent)

PROFESSIONAL EXPERIENCE

Fluid Responsible Engineer

Orlando, FL

Knights Experimental Rocketry (KXR)

January 2026 - Present

- Collaborated with multidisciplinary engineering teams to develop and integrate a fluid system for a liquid rocket engine test stand.
- Supported fluid system design through P&ID development, selecting components while balancing cost and performance, and presenting system-level designs through preliminary and critical design reviews.
- Developed MATLAB scripts to calculate mass flow rates, pressure requirements, and component sizing, using analytical models to verify fluid system performance and guide component selection.
- Modeled fluid-system components and routing in SolidWorks, integrating manifolds, tubing, regulators, and other hardware into the overall test-stand architecture.
- Developed technical documentation including system procedures, test procedures, and Concept of Operations (CONOPS) to ensure safety while operating.
- Led fluid system integration efforts following a major project redesign due to budget cuts, adapting our fluid system architecture to operate safely and effectively on an existing test stand.

WORK EXPERIENCE

Bookstore Associate

Fort Collins, CO

Colorado State University

January 2025 – March 2025

- Assisted customers and resolved inquiries through effective communication and problem-solving.
- Processed transactions with accuracy, efficiency, friendliness, and attention to detail.
- Supported inventory management, product organization, and daily store operations.
- Maintained a clean, organized, and customer-focused work environment.

PROJECTS

Self-Balancing Robot - *Personal*

- Designed and built a wheeled, self-balancing BDX-inspired droid from the ground up using SolidWorks, ESP32, and 3D printing, integrating mechanical, electrical, and embedded systems into a functional robot.
- Implemented and tuned a PID control algorithm using IMU sensor data in Arduino IDE and MATLAB to achieve autonomous balancing and to accurately control the robots' motion using a Bluetooth controller.
- Created detailed SolidWorks parts and assemblies with dimensional clearances, screw-hole tolerances, and internal component packaging to ensure proper fit and manufacturability.

L1 High-Powered Rocket - *Personal*

- Modeled and simulated a Level 1 high-power rocket in OpenRocket, evaluating multiple motor configurations, fin geometries, and nose cone designs to optimize flight performance and stability.
- Designed the rocket in SolidWorks, using CAD to determine accurate dimensions and mass properties for fabrication while applying principles of aerodynamics, propulsion, and rocket stability.